

Physical Chemistry (I)

Lecture 02



Focus vs. Chapter

Atkins	This course
Focus 1	Chapter 2
Focus 2	Chapter 1, Chapter 3
Focus 3	Chapter 4, Chapter 5
Focus 4	Chapter 6
Focus 5	Chapter 7
Focus 6	Chapter 8, Chapter 9

Thermodynamics

- Thermodynamics
 - thermo: “heat”
 - dynamics: “power”
- Study of ...
 - heat and temperature
 - and their relation to energy and work

Brief History of Thermodynamics (1)

“What is heat?”

- Heat as particles: caloric theory (17th-19th centuries)

	<i>Noms nouveaux.</i>	<i>Noms anciens correspondans.</i>
<i>Substances simples qui appartiennent aux trois règnes & qu'on peut regarder comme les élémens des corps.</i>	Lumière.....	Lumière.
	Calorique.....	Chaleur.
		Principe de la chaleur.
	Oxygène.....	Fluide igné.
		Feu.
		Matière du feu & de la chaleur.
	Azote.....	Air déphlogistiqué.
		Air empiréal.
		Air vital.
	Hydrogène.....	Base de l'air vital.
		Gaz phlogistiqué.
		Mofete.
		Base de la mofete.
		Gaz inflammable.
		Base du gaz inflammable.

(Image: <http://www.hyle.org/journal/issues/16-2/chang.htm>)

Brief History of Thermodynamics (2)

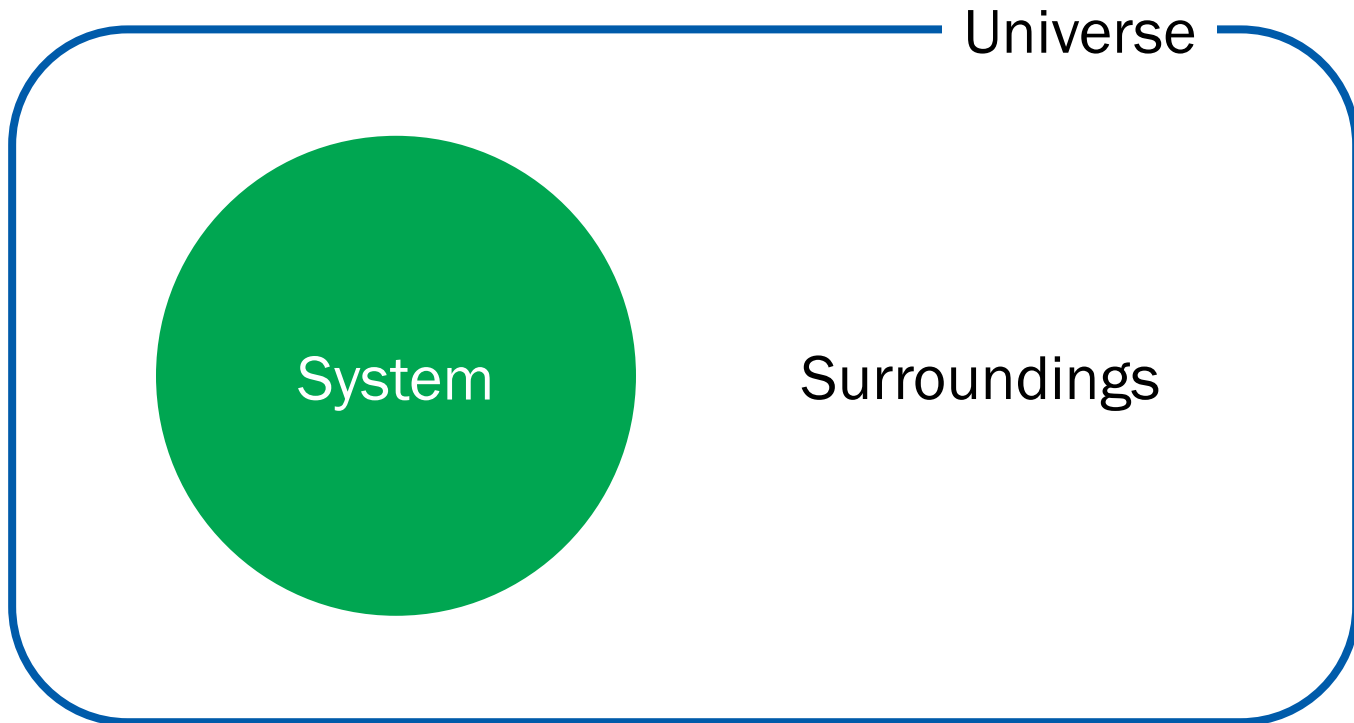
- Development of modern heat engines
 - Thomas Newcomen (1712), James Watt (1769), ...
- The mechanical value of heat: two contradicting views
 - “Fall” of a quantity of heat during the engine cycle – Carnot (1824) and Clapeyron (1834)
 - Equivalence of heat and work – Mayer (1842) and Joule (1843)
- Reconciliation: Clausius (1850) and Thomson (1851)

Brief History of Thermodynamics (3)

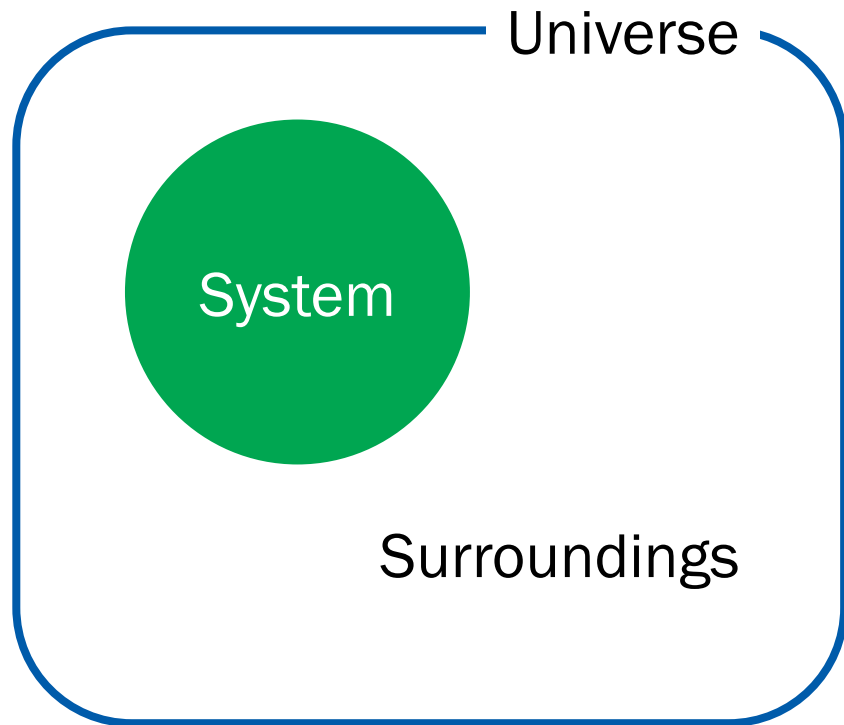
- Classical thermochemistry
 - Heat-centered view: Thomsen (1854) and Berthelot (1864)
 - Rigorous formulation: Helmholtz and Gibbs (1873-1883)
- *Zeitschrift für physikalische Chemie*
 - Founded in 1887
 - Svante Arrhenius, Wilhelm Ostwald, J. H. van't Hoff

1. Basic Concepts

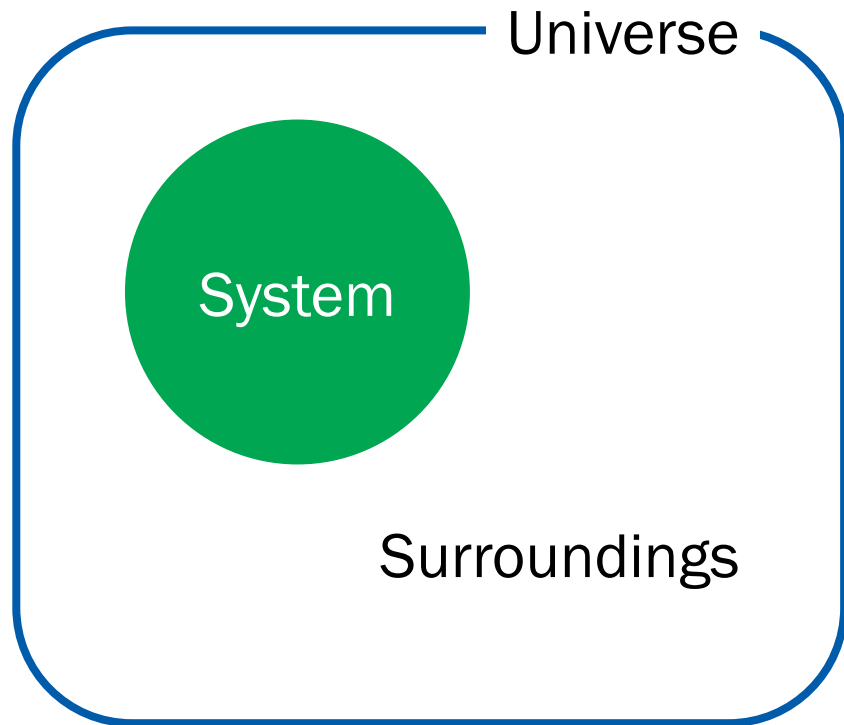
System and Surroundings



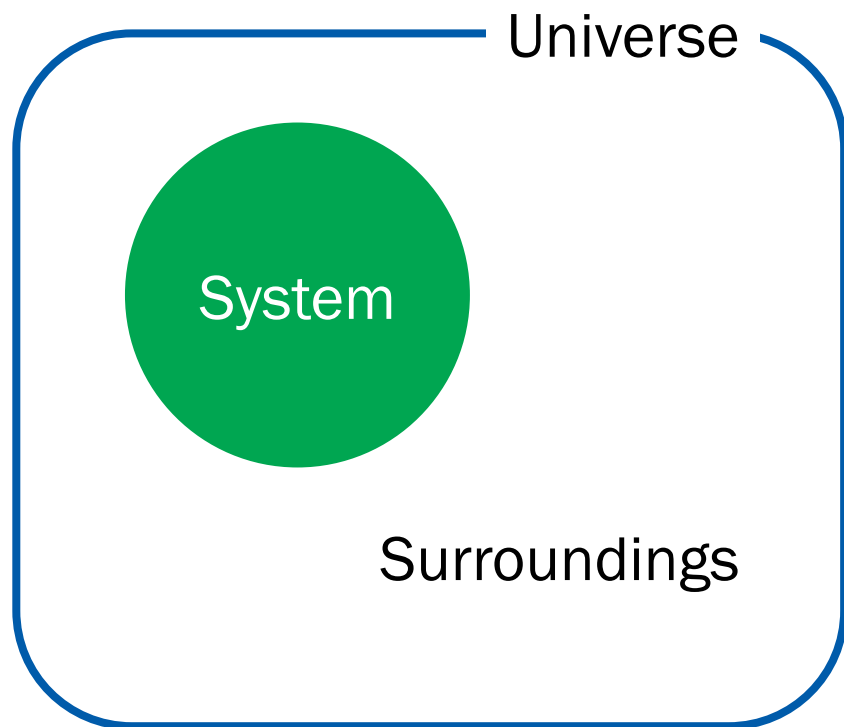
Example of a System



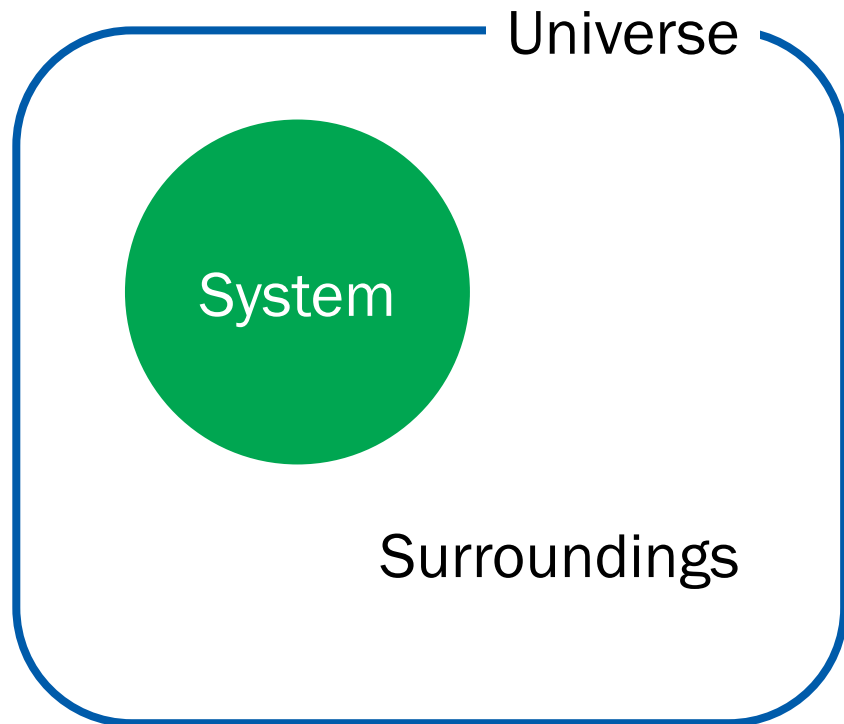
Example of a System



Another Example of a System

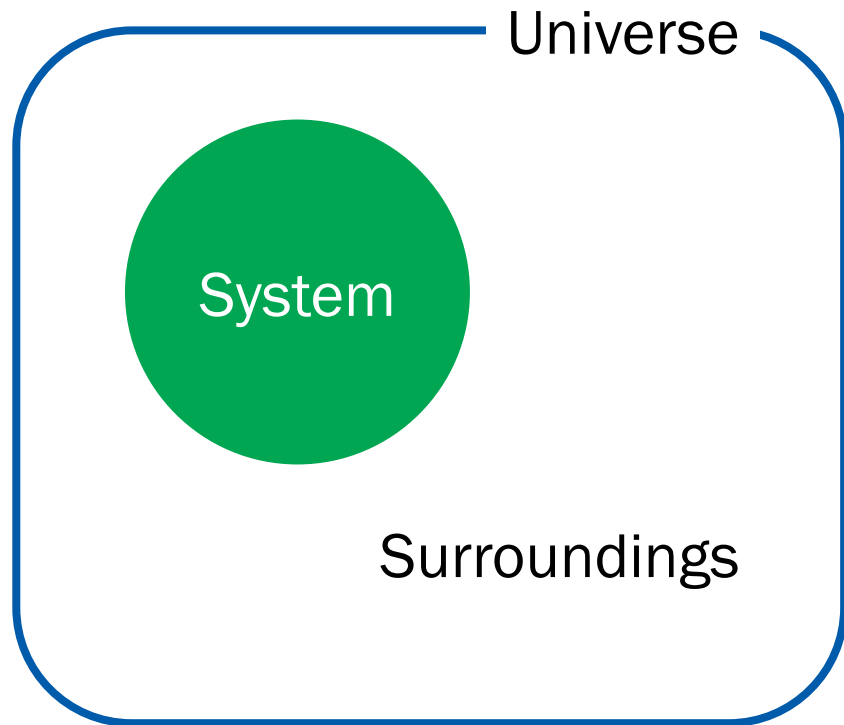


Third Example of a System

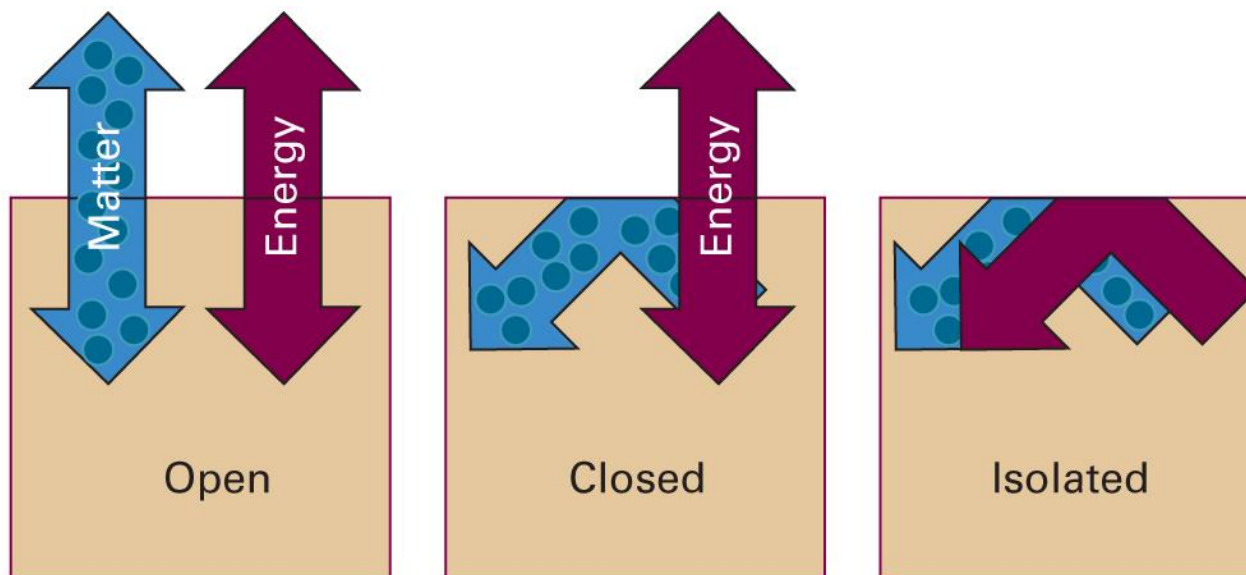


(Image: https://commons.wikimedia.org/wiki/File:Kitchen_design_at_a_store_in_NJ_refrigerator_and_cabinets_8.jpg)

And Even ...



Classification of Systems



System Properties

- A system can have different **physical properties**
- A **single value** will be assigned for each property
- Useful classification
 - **Extensive properties:** $f(A) + f(B) = f(A + B)$
 - **Intensive properties:** $f(A) + f(B) \neq f(A + B)$

Some Extensive Properties

- **Mass m :** measure of the quantity of matter present (unit: kilogram, kg)
- **Volume V :** measure of the quantity of space the sample occupies (unit: cubic meter, m³)
- **Amount of substance n :** measure of the number of specified entities (unit: mole, mol)

Relation between m and n

$$n \propto m$$

- Conversion: molar mass of a substance M (unit: kg mol^{-1})

$$n = \frac{m}{M}$$

Some Intensive Properties

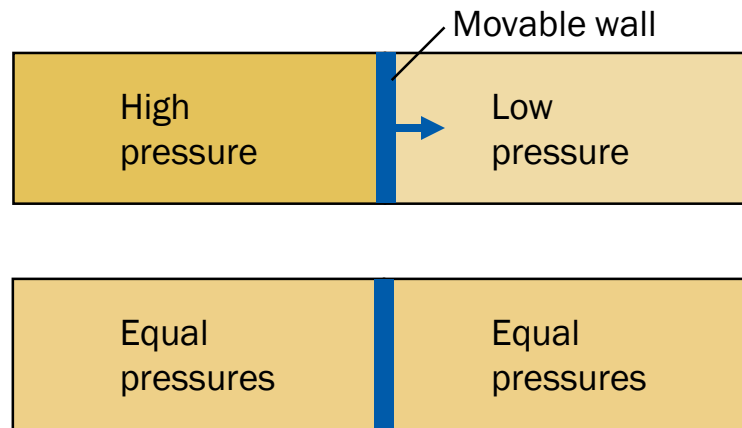
- **Pressure p :** force F / area A (unit: pascal, Pa)
 - Standard pressure $p^\ominus = 10^5 \text{ Pa} = 1 \text{ bar}$
 - Atmospheric pressure $101325 \text{ Pa} = 1 \text{ atm}$
- **Temperature T :** determines the direction of energy flow (unit: kelvin, K)
 - Celsius scale $T/\text{K} = \theta/^\circ\text{C} + 273.15$

Equilibrium

- An isolated system is **in equilibrium** when its properties remain **constant with time**
- **Equilibrium thermodynamics:** we will mostly focus on systems in equilibrium

Mechanical Equilibrium

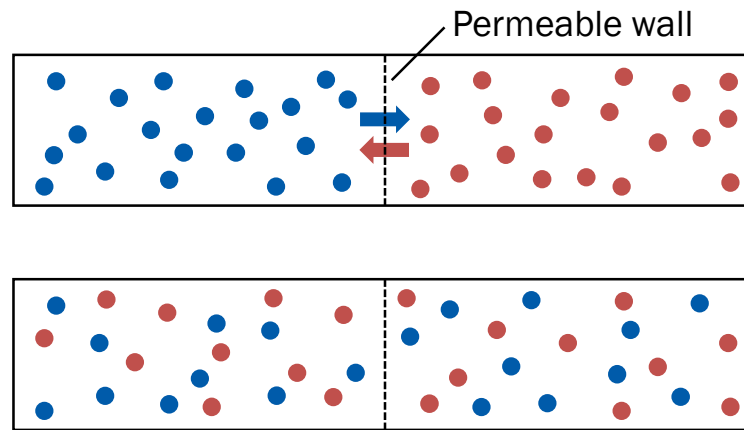
Constant p and V



Not in Atkins

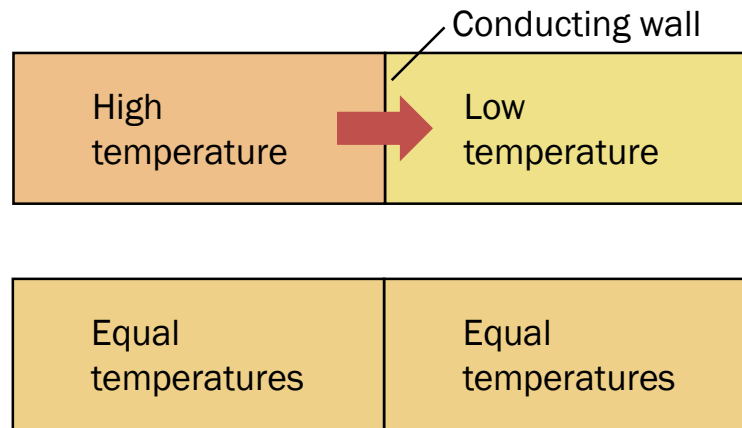
Diffusive Equilibrium

Constant n



Thermal Equilibrium

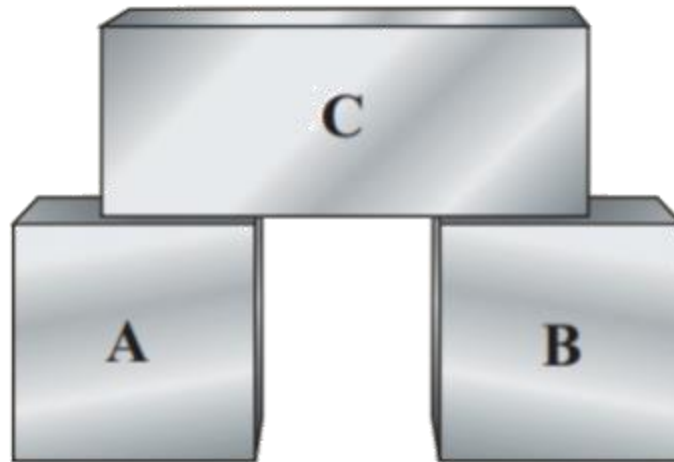
Constant T



Not in Atkins

0th Law of Thermodynamics

“If systems **A** and **B** are in thermal equilibrium, and systems **B** and **C** are in thermal equilibrium, systems **A** and **C** are in thermal equilibrium.”



(Image: http://www.brainkart.com/article/Zeroth-Law-of-Thermodynamics_36238/)

State of a System

- The **state** of a system is defined by its **properties**
- Two samples of **the same substance** that have **the same properties** are in **the same state**
- Pure substance: (n, V, p, T)

State Change (Process)

The system can change its state
from one to another

$$(n_1, V_1, p_1, T_1) \rightarrow (n_2, V_2, p_2, T_2)$$

2. The Properties of Gases

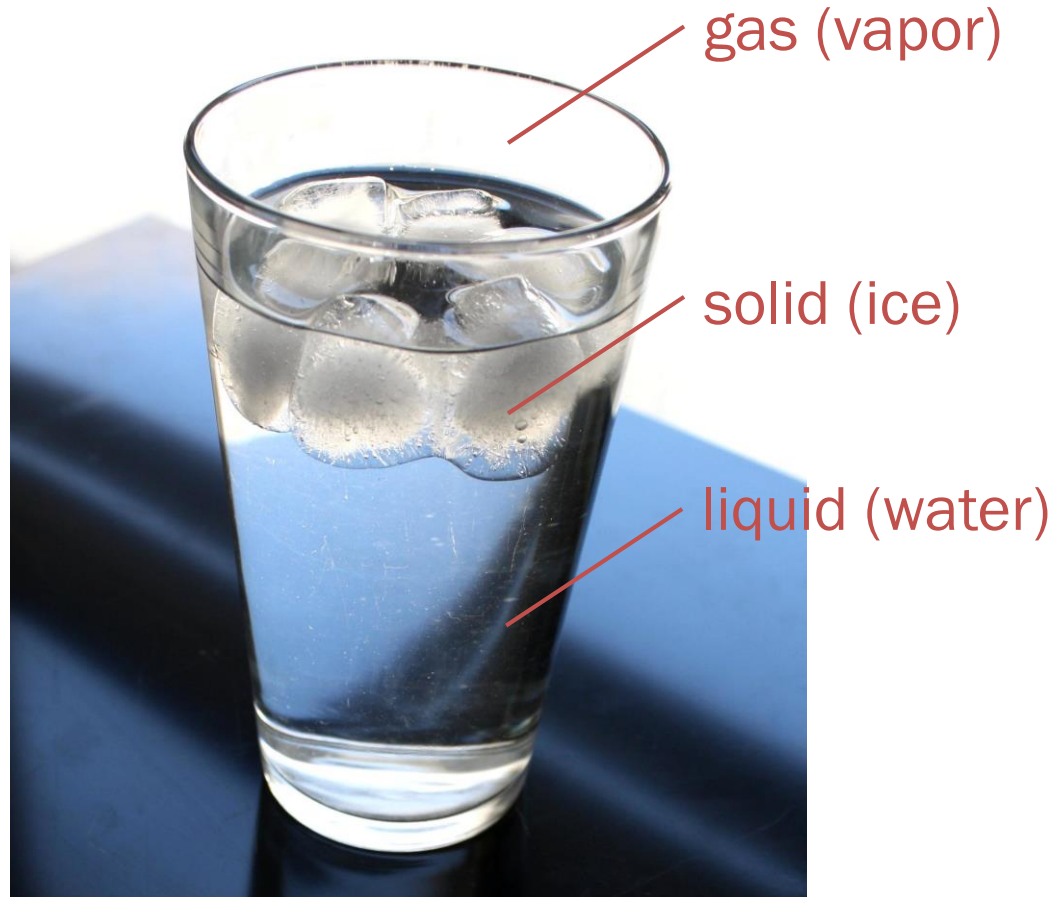


Chapter Overview

- Motivation: why gases?
- Equation of state
 - Perfect gas
 - Real gases
- Molecular kinetic theory
 - Maxwell-Boltzmann distribution
 - Collisions: molecule-molecule, molecule-wall
 - Kinetic energy of a gas

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Why Gases?

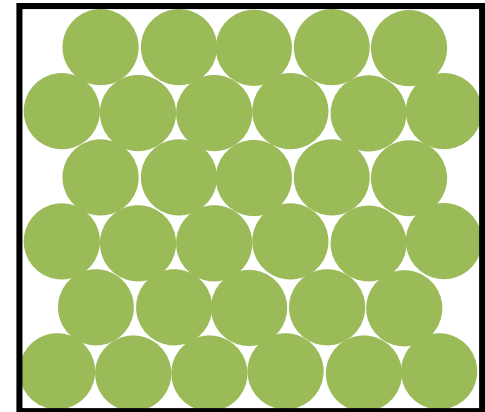
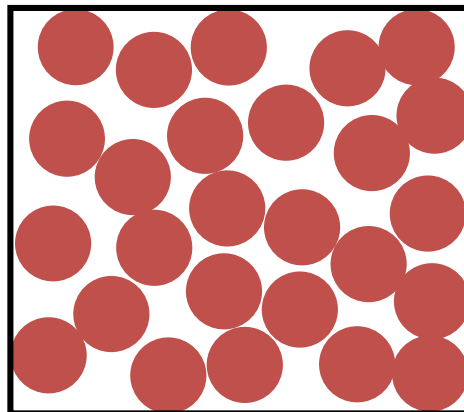
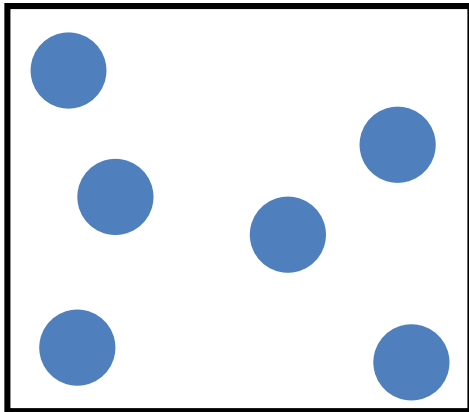


(Image: <https://pixnio.com/objects/glass/glass-ice-water>)

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Gas, Liquid, and Solid

condensed phase



Gas phase: kinetic energy $\gg$ potential energy

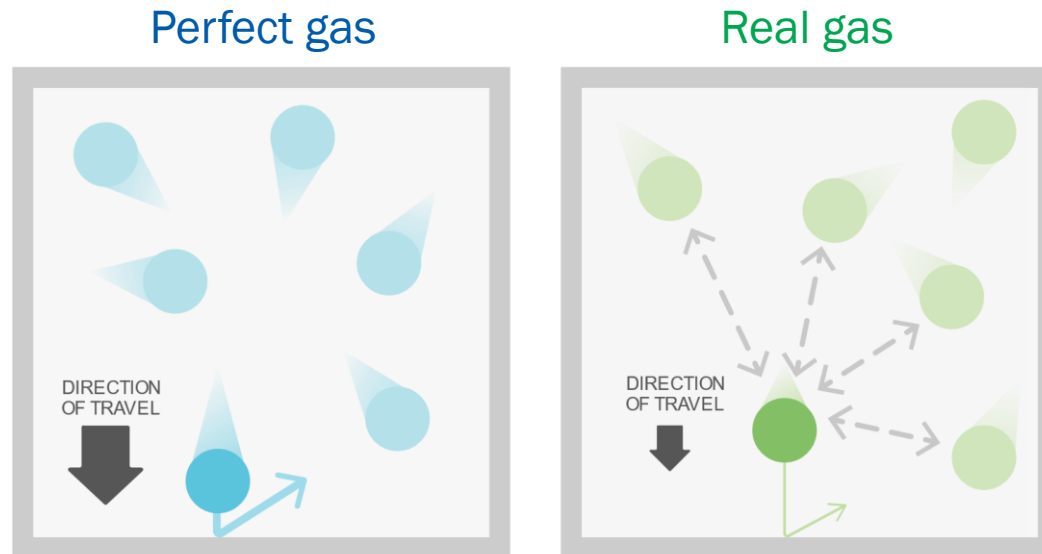
Condensed phase: kinetic energy $\approx$ potential energy

$$\frac{1}{2}mv^2$$

various forms

Perfect Gas

- Perfect gas (ideal gas): no interactions
- Real gas: finite interactions between molecules



Equation of State

- The state of a pure gas is defined by (n, V, p, T)
- **Equation of state:** relationship of the 4 variables

$$p = f(T, V, n)$$

- Equation of state of a perfect gas: **perfect gas law**

$$p = nRT/V$$

↑
gas constant

Molecular Interpretations

- Avogadro's Law: $V \propto n$

“Each molecule occupies the same space”

- Boyle's Law: $p \propto n/V$

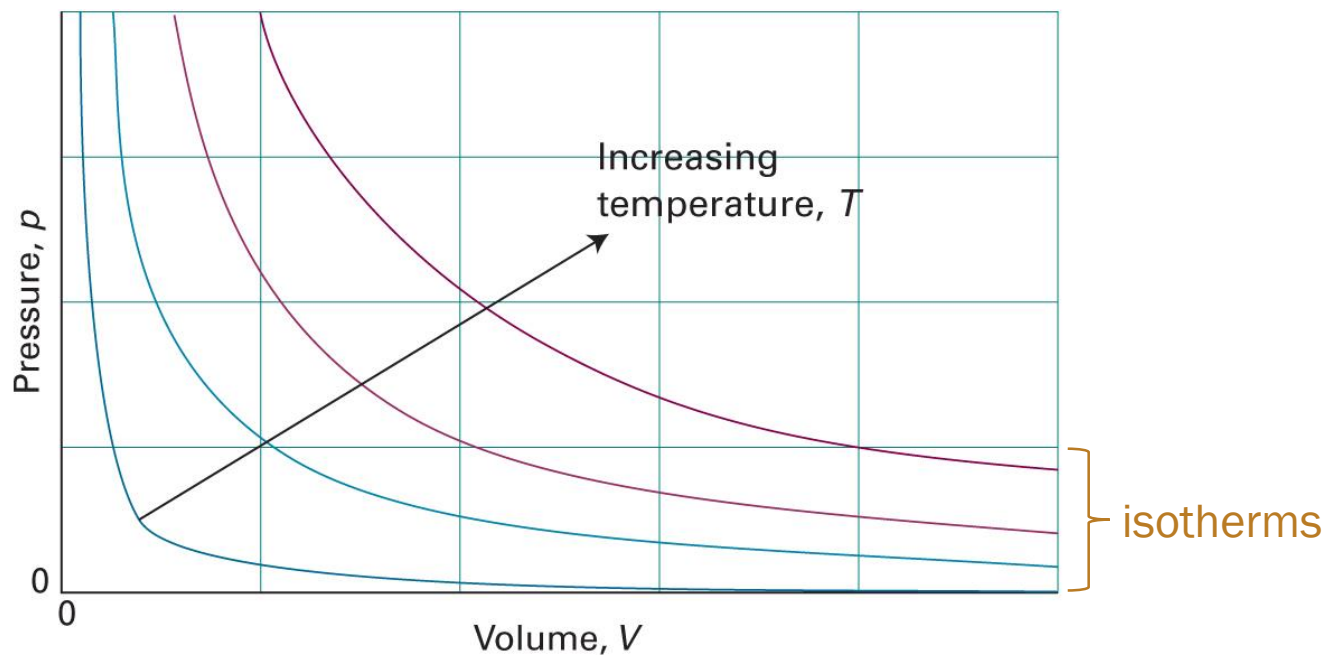
“Number of collisions is proportional to the density”

- Charles's Law: $p \propto T$

“Temperature is related to the average kinetic energy”

p - V Dependence

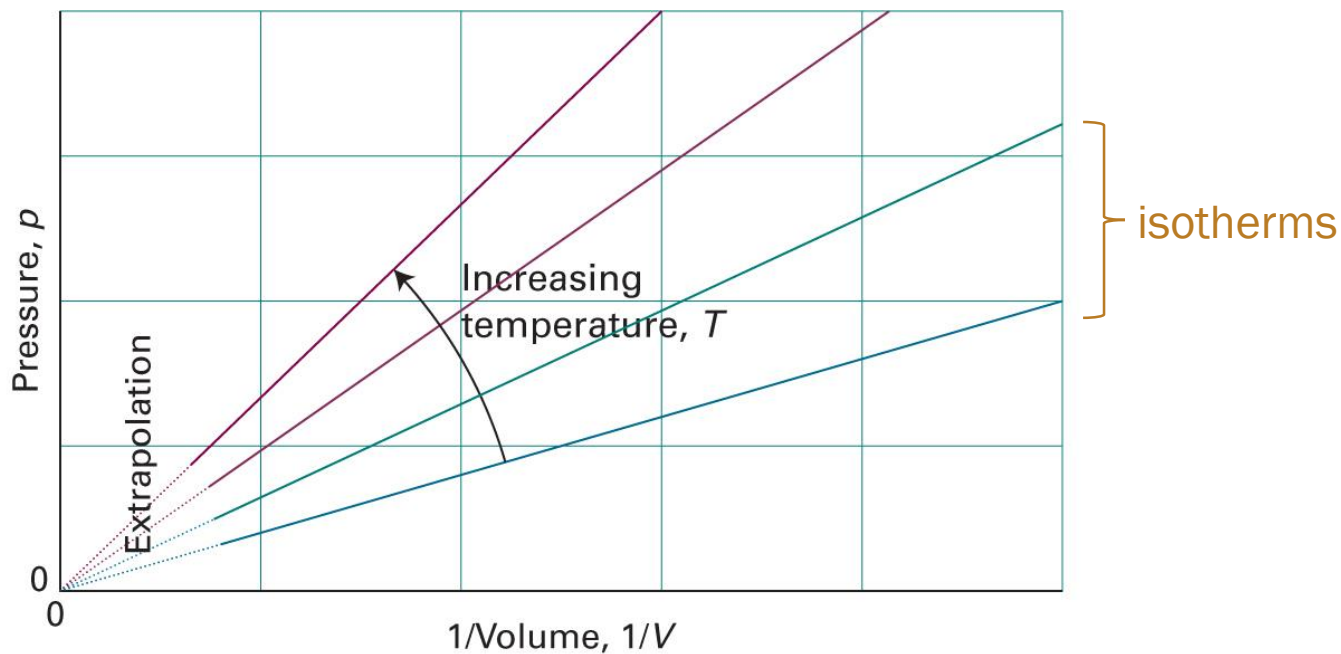
$$p = nRT/V$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 1A.2)

Another Representation

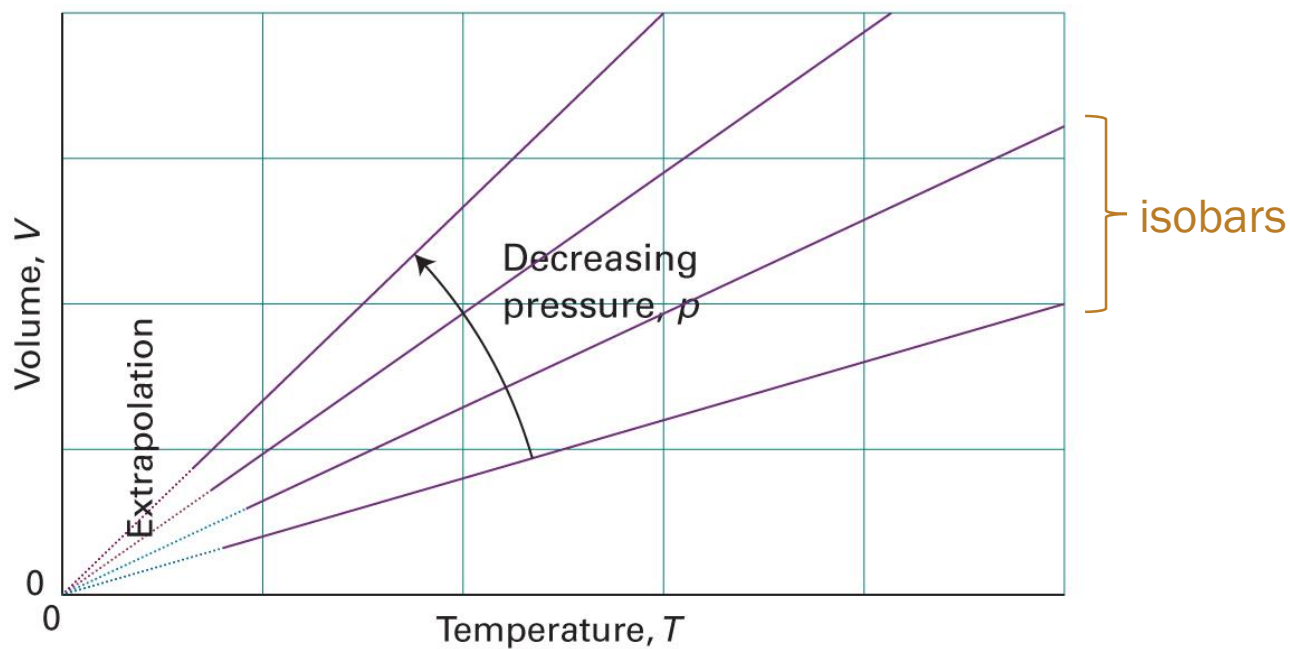
$$p = nRT/V$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 1A.3)

V - T Dependence

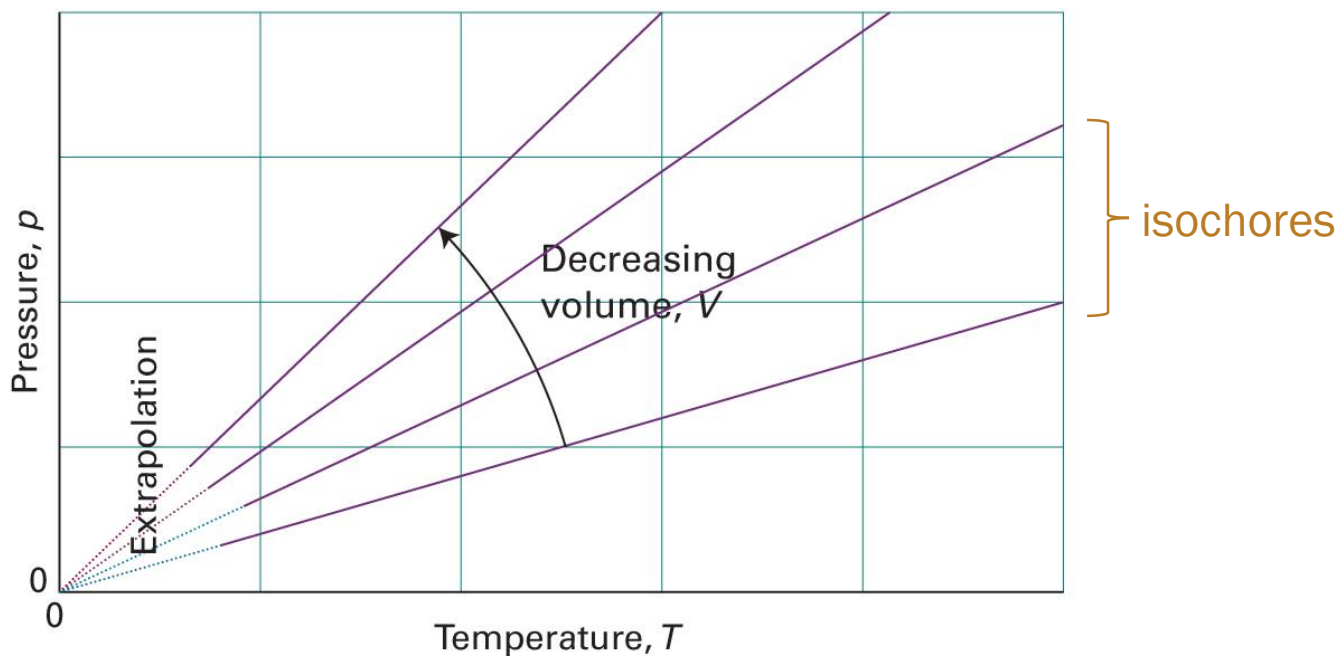
$$V = nRT/p$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 1A.4)

p - T Dependence

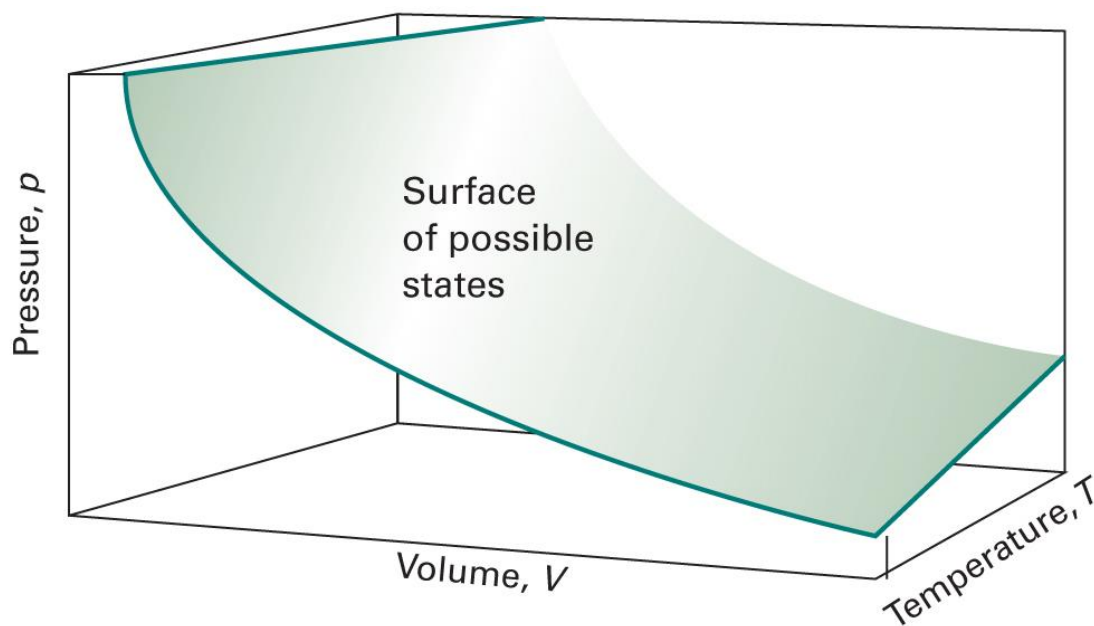
$$p = nRT/V$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 1A.5)

(p, V, T) Surface

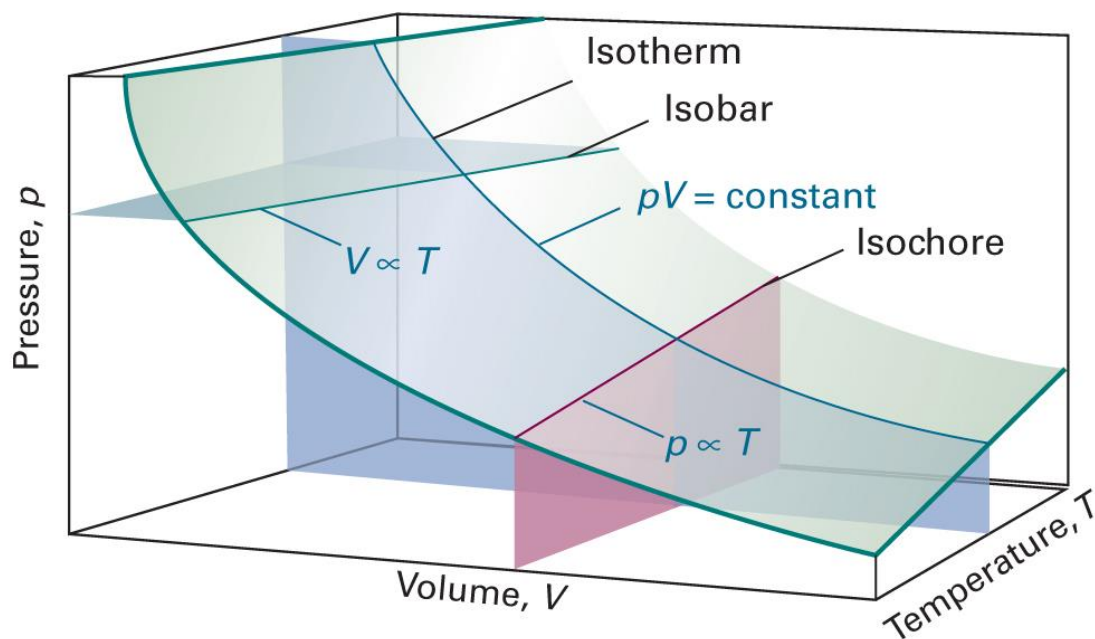
$$pV = nRT$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 1A.6)

Sections through the Surface

$$pV = nRT$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 1A.7)

Applications: Perfect Gas Law

- Limiting law: in the limit of $p \rightarrow 0$, all real gases behave as a perfect gas
- STP (standard temperature and pressure)
 - $p = 1 \text{ atm}$, $T = 0^\circ \text{C} = 273.15 \text{ K}$
 - $V/n = 22.414 \text{ L mol}^{-1}$
- SATP (standard ambient temperature and pressure)
 - $p = 10^5 \text{ Pa}$, $T = 25^\circ \text{C} = 298.15 \text{ K}$
 - $V/n = 24.789 \text{ L mol}^{-1}$

Combined Gas Law

From the perfect gas law,

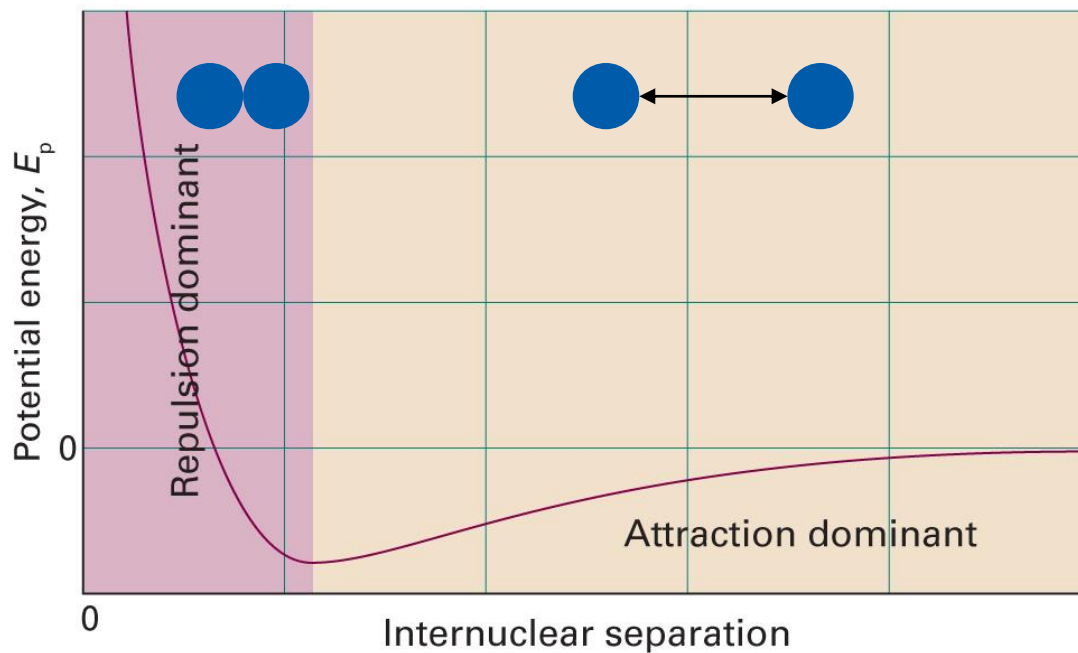
$$R = \frac{pV}{nT} = \text{constant}$$

Hence, for a change from state 1 to state 2,

$$\frac{p_1 V_1}{n_1 T_1} = \frac{p_2 V_2}{n_2 T_2}$$

Real Gas

Now the molecules interact!



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 1C.1)

Compression Factor Z

$$Z = V_m/V_m^0$$

Molar volume of a gas $V_m = V/n$

Molar volume of a perfect gas V_m^0

Since $V_m^0 = RT/p$,

$$Z = \frac{pV_m}{RT} = \frac{pV}{nRT}$$

Amplitude of Z

$$Z = V_m/V_m^0$$

- $Z > 1$:

$V_m > V_m^0 \rightarrow$ repulsion is dominant

- $Z \approx 1$:

$V_m \approx V_m^0 \rightarrow$ perfect-gas-like behavior

- $Z < 1$:

$V_m < V_m^0 \rightarrow$ attraction is dominant

Brief Illustration 1C.1

The molar volume of a perfect gas at 500 K and 100 bar is $V_m^0 = 0.416$ L mol⁻¹. The molar volume of carbon dioxide under the same conditions is $V_m = 0.366$ L mol⁻¹. What is the compression factor?

$$Z = V_m/V_m^0 = (0.366 \text{ L mol}^{-1})/(0.416 \text{ L mol}^{-1}) = 0.880$$